

LANBridge — Developer Requirements Document

Virtual LAN Gaming Infrastructure for LAN Multiplayer Games

1. Project Overview

Project Name

LANBridge

Project Type

Windows Desktop Networking Application

Project Category

Virtual LAN / Gaming VPN Infrastructure System

Purpose of the Project

LANBridge is a lightweight virtual LAN application designed to allow users in different physical locations to play LAN-based multiplayer games together over the internet.

The application should simulate a local area network (LAN) environment between remote systems so that games which normally require the same Wi-Fi or same router can discover and connect to each other.

The goal is NOT to modify games.

The goal is to create a virtual network tunnel that makes remote users appear to be on the same local network.

2. Main Problem Statement

Many classic multiplayer games only support LAN connectivity.

Examples:

- Need for Speed Underground 2
- Stronghold Crusader
- Counter-Strike 1.6

These games normally require:

- same router
- same Wi-Fi
- same local subnet

Users located in different cities or different networks cannot normally discover each other inside the LAN game lobby.

LANBridge will solve this problem by:

- creating a virtual LAN tunnel
 - assigning virtual IP addresses
 - forwarding LAN packets through the internet
 - making remote systems behave like same-network systems
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3. Target Platform

Operating System

Windows 10 and Windows 11

Architecture

64-bit desktop application

Application Type

Client-server desktop networking application

4. Core Objectives

The application must:

1. Allow users in different locations to connect together.
 2. Create a virtual local area network.
 3. Assign virtual IP addresses.
 4. Forward multiplayer game packets.
 5. Allow LAN-based games to discover each other.
 6. Minimize latency for multiplayer gameplay.
 7. Provide a simple and lightweight interface.
 8. Support multiplayer testing with:
 - Need for Speed Underground 2
 - Stronghold Crusader
 - Counter-Strike 1.6
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5. Project Scope

This project is intended as:

- an academic networking infrastructure project
- a lightweight gaming VPN
- a LAN emulator system
- a multiplayer tunneling application

This project is NOT intended to:

- replace commercial VPN systems
 - provide military-grade security
 - support all online games
 - bypass anti-cheat systems
 - modify game files
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6. Expected Workflow

Scenario Without LANBridge

User A:

- connected to mobile hotspot A

User B:

- connected to mobile hotspot B

Result:

- games cannot discover each other
 - LAN multiplayer fails
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Scenario With LANBridge

Step 1:

Users open LANBridge.

Step 2:

Users join same virtual room.

Step 3:

LANBridge creates a virtual network tunnel.

Step 4:

Application assigns virtual IP addresses.

Step 5:

Game LAN traffic is forwarded through tunnel.

Step 6:

Games detect users as if they are on same Wi-Fi.

Result:

- LAN lobby becomes visible
 - multiplayer works remotely
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7. Functional Requirements

7.1 Room Connection Module

Features

- Create room
- Join room
- Display connected users
- Show host IP
- Show virtual IP

Requirements

- No account system required
 - No registration required
 - No passwords required
 - Connection should work using room name or host IP only
 - Keep implementation lightweight and simple
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7.2 Virtual Network Module

Purpose

Create virtual LAN between connected systems.

Requirements

- Create virtual IP assignment system
- Maintain virtual subnet
- Simulate local network environment

Example

User A:
10.0.0.2

User B:
10.0.0.3

7.3 Packet Forwarding Module

Purpose

Forward multiplayer packets through internet.

Requirements

- Capture LAN traffic
- Tunnel packets
- Forward traffic between connected peers
- Support UDP communication
- Support TCP if needed

Important

UDP should be prioritized because multiplayer games require low latency.

7.4 Room System

Features

- Create room
- Join room
- Leave room
- Show room members

Requirements

- Lightweight implementation
 - Real-time updates
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7.5 Connection Monitoring Module

Features

- Show ping
- Show connected players
- Show room status
- Show virtual IP
- Show connection state

UI Requirement

Very simple interface.

No advanced dashboard required.

7.6 Reconnection System

Requirements

- Detect disconnection
 - Auto reconnect if internet briefly disconnects
 - Preserve room session
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8. Non-Functional Requirements

Performance

- Lightweight
- Low CPU usage
- Low RAM usage
- Low latency communication

Reliability

- Stable tunnel connection
- Reconnection support

Compatibility

- Windows 10
- Windows 11

Usability

- Simple UI
 - Beginner-friendly operation
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9. Recommended Technologies

Backend

Recommended:

- GoLang
OR
- Node.js
OR
- Python

Preferred:

GoLang because of:

- networking performance

- concurrency support
 - lightweight deployment
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Frontend

Recommended:

- Electron.js
OR
- Tauri
OR
- Simple Python GUI

Requirement:

Minimal interface only.

Networking

Required:

- UDP sockets
 - TCP sockets
 - packet forwarding
 - tunneling logic
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Database

Recommended:

- SQLite

Reason:

Simple and lightweight.

10. Networking Concepts Used

The application should implement or simulate:

- Virtual LAN
 - VPN tunneling
 - UDP communication
 - TCP/IP networking
 - Packet forwarding
 - Socket programming
 - Virtual IP assignment
 - Network encapsulation
 - Peer communication
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11. Supported Games for Testing

Primary Testing Games

Need for Speed Underground 2

Purpose:

Main gameplay demonstration.

Stronghold Crusader

Purpose:

Strategy game LAN demonstration.

Counter-Strike 1.6

Purpose:

Stable networking test environment.

12. UI Requirements

Main Screen

The application should contain:

- Create room button

- Create room button
 - Join room button
 - Host IP display
 - Connected users list
 - Ping display
 - Virtual IP display
 - Connection status
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UI Design Rules

- Dark theme preferred
 - Lightweight interface
 - Minimal animations
 - Fast loading
 - Simple navigation
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13. Suggested Architecture

Client-Server Hybrid Architecture

Central Server Responsibilities

- room coordination
- room management
- connection coordination

Client Responsibilities

- packet forwarding
 - game communication
 - LAN simulation
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14. Suggested Development Phases

Phase 1 — Research

Tasks:

- Research LAN discovery methods
 - Study UDP communication
 - Study VPN tunneling basics
 - Analyze multiplayer behavior of selected games
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Phase 2 — Core Networking

Tasks:

- Build socket communication
 - Create room connection system
 - Establish peer communication
 - Test packet transfer
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Phase 3 — Virtual LAN System

Tasks:

- Assign virtual IPs
 - Simulate local network
 - Forward LAN packets
 - Implement basic tunnel system
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Phase 4 — Game Testing

Tasks:

- Test Counter-Strike 1.6
 - Test Stronghold Crusader
 - Test Need for Speed Underground 2
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Phase 5 — UI Development

Tasks:

- Create desktop interface
- Add room management
- Add connection status indicators

Phase 6 — Optimization

Tasks:

- Reduce latency
 - Improve reconnection handling
 - Improve tunnel stability
 - Optimize packet handling
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15. Basic Technical Flow

Example Workflow

1. User launches application.
 2. User logs into LANBridge.
 3. User creates or joins room.
 4. Server coordinates room connection.
 5. Virtual IP assigned.
 6. Tunnel established.
 7. LAN packets forwarded.
 8. Game detects remote player.
 9. Multiplayer session starts.
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16. Important Implementation Notes

Priority

The project should focus on:

- stable LAN discovery
- low latency
- reliable packet forwarding
- successful multiplayer detection

NOT on:

- advanced graphics

- advanced AI
 - unnecessary animations
 - complex enterprise features
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17. Demo Requirements

The final demonstration should show:

Step 1

Two systems connected to DIFFERENT hotspots.

Result:

Games cannot discover each other.

Step 2

Enable LANBridge.

Step 3

Join same virtual room.

Step 4

Open multiplayer game.

Result:

LAN lobby becomes visible.

Step 5

Start gameplay successfully.

18. Success Criteria

The project will be considered successful if:

- remote systems can discover each other
 - games detect virtual LAN correctly
 - multiplayer session starts successfully
 - gameplay remains stable
 - latency remains acceptable
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19. Risks and Limitations

Possible limitations:

- Windows firewall issues
 - NAT restrictions
 - packet loss
 - old game compatibility issues
 - internet quality dependency
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20. Final Deliverables

The final project should include:

- Windows desktop application
 - server application
 - networking modules
 - room system
 - multiplayer tunnel system
 - demonstration setup
 - project documentation
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21. Final Goal

The final goal of LANBridge is to create a lightweight virtual LAN gaming infrastructure system that allows classic LAN multiplayer games to work across remote internet connections while maintaining a simple and stable user experience.